

Mouse auditory experiment

A first plan

October 8, 2026

We want to play short tones to a mouse and record its brain responses. SONIC tested awake sheep; our mouse state and tone settings remain open. (Methods 6.6, p.13)

1. Choose the sounds

- Choose the frequency, duration, loudness, and gap between tones.
- **SONIC:** initially explored 1.4–32 kHz; benchmark tones were 10 ms sine waves with 2 ms fades at each end. (Methods 6.1–6.2, p.11)
- **SONIC:** random frequencies, balanced within each block, with no silence between tones. (Methods 6.1, p.11)
- Save the sequence before playback, as SONIC did. (Methods 6.2, p.11)

2. Set up the speaker

- Plan as though we need to assemble the whole sound setup.
- **SONIC:** NI USB-6361 output at 192 kHz, TDT SA1 amplifier, and MF1 free-field speaker; about 80 dB SPL at sheep-ear distance. (Methods 6.2, p.11)
- **Our plan:** use the proposed shopping list on the next page, keep speaker placement fixed, and measure each tone with a calibrated microphone at the mouse-ear position.

3. Keep everything on one timeline

- Schedule each tone with a timing pulse and record it alongside the neural signals.
- **SONIC:** recorded electrical sound output and trial-start markers; shared clock references aligned these with Connexus brain recordings. (Methods 6.2, p.11)
- **Our additions:** synchronize a microphone recording to check actual sound arrival. Align video frame times too if we add behavior video.

Reference: Perkins et al. (2025), SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces, bioRxiv v1. Page numbers above refer to the printed manuscript. Mouse choices and additions are proposals.

One proposed shopping list

Building the mouse setup from scratch

USD

Budget for one of each item or group below. **Est.** means our rough allowance, not a confirmed supplier price.

What we get	What it does	Budget
1 TDT MF2-M speaker kit	Plays the tones.	\$1,500 est.
1 TDT SA1 amplifier	Drives the speaker.	\$2,000 est.
1 TDT ZB1PS power unit	Powers the amplifier.	\$750 est.
1 NI USB-6361	Sends tones and timing signals.	\$3,030
1 complete Avisoft 116Hm kit	Records sound at the ear position.	\$8,160
Calibration support	Checks the sound levels.	\$1,500 est.
1 Intan recording set	Controller, 32-channel headstage, and cable for brain signals.	\$11,105
1 electrode and adapter	Connects the brain recording to the headstage.	\$1,500 est.
1 computer, monitor, and storage set	Runs the setup; includes a backup drive.	\$2,000 est.
Cables, timing connections, and mounts	Connects equipment and fixes its position.	\$1,000 est.
Equipment subtotal	Rounded budget	\$33,000

What is already included?

- The Avisoft kit includes the microphone, preamp, sound recorder, cable, stand, and software. Buy **one kit**.
- The MF2-M kit includes a stand and connection accessories. No extra speaker or microphone is needed for this budget.

What could add to the cost?

- Allow **\$5,000 extra** for animal support equipment; the mouse preparation is still undecided.
- Keep **\$4,000 reserve** for price changes, shipping, and tax: working budget **about \$42,000**.
- Optional behavior camera with accessories: **about \$1,100 extra**.

Before buying: Intan is a budget example. The microphone's stated band starts at 2 kHz; lower tones need a different choice. Calibration and timing need checking. Animal support excludes housing, full surgery/anesthesia infrastructure, and ongoing costs. A soundproof box is not required.

Sources, October 8, 2026: TDT kit, amplifier, power, NI, Avisoft kit, prices, Intan. SONIC used MF1; MF2 is our current candidate. Estimates/alternatives are detailed in the repository. This mouse draft is separate from Rice's sedated-sheep plan.